

Mitigating Dataset Artifacts Derived from Learning an NLI Task

Authors

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Abstract

The purpose of this paper is to explore a situation where a model learns dataset artifacts found in the SNLI dataset, using ELECTRA-small as the model used for the NLI classification task. Furthermore, we use Facebook’s ANLI dataset as supporting evidence to prove that the model learned dataset artifacts instead. After proving the existence of these artifacts, we then try to stack two approaches to fix the model’s performance. For the first approach, we trained the model on the adversarial dataset, ANLI so that the model can better learn the task from more complicated data. After seeing an improvement in evaluation accuracy, we curate contrastive sets to evaluate the model’s weaknesses and add new training examples to attempt and improve the accuracy of the ANLI based model. Doing so helped the model further separate the embedding spaces of each class, which enabled it to make more accurate predictions.

1 Introduction

There are certain conditions for when a model gives the illusion that it is learning a task but doesn’t learn how to do it. It is even able to derive decent evaluation performance, which further gives the impression that it’s learning to do said task. However, the model isn’t learning the right objective, instead it’s learning patterns in the dataset that allow it to output decent performance on the in-distribution evaluation set. For example, while training on a NLI classification task, the model could have noticed samples which contained the word “cat” only had negative labels

assigned to them, which means that any new sample that contains the word “cat” would also be classified as negative. Patterns or correlations such as these are referred to as dataset artifacts. In this paper, we’ll be further exploring this phenomenon and attempting to mitigate it so our model can learn the actual task at hand.

1.1 The Dangers of Learning Dataset Artifacts

When the model learns dataset artifacts instead of the task, a problem arises: the model isn’t capable of being generalized to out-of-distribution datasets.

To prove this, we fine tune the ELECTRAsmall (Clark et al., 2020a) model on the SNLI dataset (Bowman et al., 2015) for the purpose of natural language inference classification. Upon fine tuning ELECTRA-small with the SNLI dataset, we then use the in-distribution evaluation set to see the model’s ability to generalize.

The results on the evaluation set seem promising and give the impression that the model is successfully learning NLI classification. The evaluation metrics are listed below.

Label	Precision	Recall	F1- Score
0	0.91	0.91	0.91
1	0.86	0.86	0.86
2	0.92	0.91	0.91
Accuracy			0.90

Figure 1: SNLI Evaluation Metrics

To confidently determine if a trained model has learned its assigned task, researchers have developed different methods of verification. One of these approaches is denoted as adversarial sets (Liu et al., 2019; Zhou and Bansal, 2020; Morris et al., 2020). In our analysis, we use an adversarial set to evaluate whether fine tuning the ELECTRAsmall model on the SNLI dataset definitively allowed the model to learn natural language inference classification.

The selected adversarial set is the ANLI dataset (Nie et al., 2020), curated by Facebook, which stands for “Adversarial Natural Language Inference.” We did not train the model on this dataset but instead used ANLI only as our evaluation set. Upon seeing the results of the evaluation, we can confidently conclude that the model did not actually learn the NLI classification task, but instead learned dataset artifacts present in the SNLI dataset. The evaluation metrics are listed below.

Label	Precision	Recall	F1-Score
0	0.36	0.36	0.36
1	0.31	0.38	0.34
2	0.31	0.24	0.27
Accuracy			0.33

Figure 2: ANLI Evaluation Metrics

The metrics above clearly outline the model’s failure to generalize itself to other datasets, which enables us to conclude that our trained model had done well on the in-distribution evaluation set due to learning dataset artifacts present in SNLI.

While giving the initial impression that the model had learned the task, we soon find out that it hadn’t after evaluating the model on ANLI. This illusion can lead to detrimental consequences if deployed to a production environment, thus it is important to stress test these trained models to ensure that they can generalize to many types of valid data samples.

1.2 Analyzing the Model’s General Mistakes

We now perform error analysis on the ANLI dataset and attempt to correlate it to potential dataset artifacts that the model could have learned.

We start by analyzing the confusion matrix which compares the model’s outputs with the true labels.

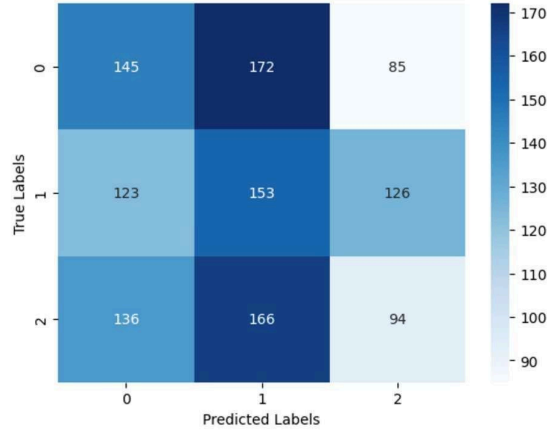


Figure 3: Confusion Matrix after ANLI Evaluation

From inspecting the confusion matrix, we see that the ELECTRA-small model struggles to correctly predict contradiction samples. The model tends to predict “neutral” and even “entailment” more than “contradiction” for samples that had “contradiction” as it’s label.

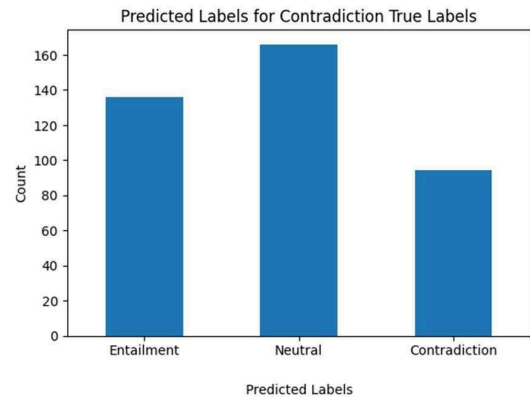


Figure 4: Predicted Label Counts for Samples with Contradiction True Label

Furthermore, the model cannot accurately predict the neutral label for most neutral samples as well. Its predictions for neutral samples seem to be split across all labels equally, which implies

that the model is also struggling to differentiate what a neutral sample looks like.

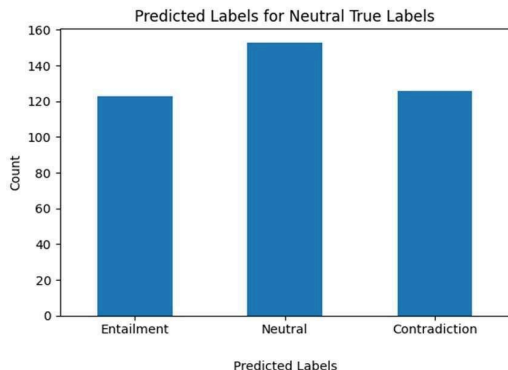


Figure 5: Predicted Label Count for Samples with Neutral True Label

From the confusion matrix we see that the model was not able to accurately predict most entailment examples either, but it did a better job for entailment compared to the other samples.

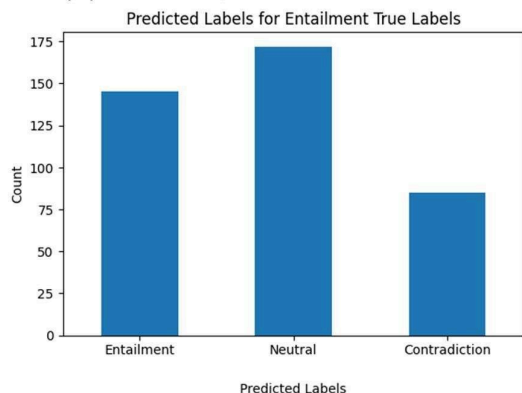


Figure 6: Predicted Label Count for Samples with Entailment True Label

In general, the model struggled to differentiate between the label classes, showing particularly poor performance on contradiction samples. This indicates that the model failed to effectively separate the embedding spaces of the classes, with potential overlap between the spaces themselves. In a further section ahead, we attempt to separate the embedding spaces using contrastive sets (Dua et al., 2021).

Furthermore, we looked at incorrect predictions where the model was “not confident” with its

choice and got it wrong. We define “not confident” as having a max prediction score that evaluates to a percentage less than 50%, meaning the model was less than 50% confident with its answer. This implies that the model almost chose the other options. We tried to understand if there were similar properties about those samples that made the model unsure of the correct label. The error class we derived consists of not understanding one or two key words in the hypothesis or premise. For example, the model was “not confident” with one of its predictions, we hypothesize, because it does not understand the difference between Reuters and NY Times. Another sample is listed below.

Hypothesis	Premise	Label	Predicted Label
... The sender claimed to have had an associate plant a small bomb in the recipient's building and said the only way to stop him from setting it off was by making an online payment of \$20,000 in bitcoin.	The sender would have accepted \$20,000 in any crypto currency.	Contradiction	Entailment (Crypto vs Bitcoin)

Figure 7: Example of Model Not Understanding Key Words (Crypto vs. Bitcoin)

We’ve identified 96 samples of these “not confident” predictions after evaluating on the test_r3 split from ANLI. Some other key words we believe the model did not understand were “Disney World”, “subjective”, “sent in”, “intersection”, etc.

Upon further investigation of the “notconfident” samples, we noticed several other categories that contributed to a negative influence towards the model’s accuracy. The first category is having specific context or domain knowledge around the premise. Several examples exhibited a hypothesis that required previous knowledge within the

domain so that the model can correctly articulate if the hypothesis can be inferred from the premise. This does raise the question that the model is capable of differentiating between the different types of labels but should it be restricted to certain domains and require that the premise contains sufficient knowledge for a hypothesis or should the neutral label indicate a model’s inability to apply the hypothesis to the premise due to lack of knowledge.

The second category we noticed is numerical inference for either a cumulation of objects or a mixture of numbers that can easily throw off the model given a hypothesis with context focused around such values. One example we noticed was that the hypothesis asked if a certain count of people was within the “100s”, denoting to us, the readers, that the hypothesis aims to find a number between 100 to 999 but to the model, it predicted neutral indicating that it miss-understood what the hypothesis was aiming for. This does indeed lead to the inability to comprehend an exact value match such as dates or numbers without a clear direction within the hypothesis.

The third category is similar to the first category but instead of domain knowledge it’s more about characteristics or features of a subject or an object. Some of the hypotheses assumed certain aspects of a character that seemed to not exist anywhere within the premise which makes it difficult for the model to learn about the character and in general make a global assumption that every character has the possibility of having such feature. This not only generalizes the model to assume that such hypotheses are common but it may also influence its future decision when comparing such features with the natural aspect of the character leading it to assume that a hypothesis could theoretically be true just because an example proceeded it to be true that contain that feature description.

Lastly, it is important to acknowledge that there are many more types of categories that the model may have generalized which will produce artifacts when exposed to new data. With our research on the mistakes the model made, we were able to find several examples that fit each category described above and have made attempts to resolve them by fine-tuning the model on data that counter or assist in correctly predicting the outcome that makes the most sense given the minimal amount of data. The

next section will discuss the actions taken and what results were yielded from this fine-tuning.

2 Improving the Model

In the previous section, we saw that the model was not capable of understanding the nuances between each of the label classes. In other words, it did not fully separate the embedding spaces enough of the label classes to distinctly identify one label over another. As Figure 4 showed, it got most of the samples labeled as contradiction incorrect. Figures 5 and 6 showed that the model almost evenly predicted all labels for the other classes. In this section, we try stacking different approaches to improve the model’s original evaluation accuracy of 33% on the ANLI test_r3 split.

2.1 Training ELECTRA-SMALL on ANLI

For the first approach, we attempt to train ELECTRA-small on the adversarial dataset, ANLI, that we used in the previous section. By the nature of adversarial datasets, their intent is to verify that the model learned the task by introducing the model to more complicated data (Liu et al., 2019; Zhou and Bansal, 2020; Morris et al., 2020). Therefore, we trained the model on the train_r3 split from ANLI so that it can be exposed to more complex examples that will allow the model to find more patterns and create stronger associations. The intent is that the model will, as a result, better separate the embedding spaces of the classes.

2.2 Using Contrastive Sets

The next iteration of fine-tuning the model to improve its accuracy was to experiment with Contrastive Sets. These types of sets are used on the model to evaluate its performance on a group of either extremely difficult cases or a presumed edge case that may be of interest (Dua et al., 2021). In the case of this research paper, we have already identified the three categories in which the model has exhibited a weakness in accuracy as discussed above in section 1.2. The main premise with this approach is to gain a better understanding in where the model may lack examples and then of course re-train it to improve itself on such examples.

On a high level, these sets as described contain examples that tackle certain problems. Given that the original model was trained on at least 100,000 different examples from ANLI/train_r3, it would be difficult to manually go through every example and evaluate the model's performance based on that. Instead, we relied on the model's initial evaluation and its accuracy when it comes to its prediction vs. the gold label and extracted the incorrectly labeled examples to use as a basis for our Contrastive Set. We examined the premise and the hypothesis and manually assessed if the gold label made sense given solely those two contexts, this way if the model was very confident in its incorrect label, we can then presume that it either does not have sufficient domain knowledge or a weakness to those certain types of passages.

Our first step was to randomly capture examples that the model predicted incorrectly. We decided to choose 7 examples to refine that fit in the categories discussed previously. After gathering the examples, we started to manually examine each one and fed it through the model to be evaluated by itself. This gave us more control to see how the model is behaving when we change certain terminology within the hypothesis. For some examples we repeated this several times depending on how complex and inaccurate the model's scores are. Then, we fed the same premise and hypothesis into ChatGPT (Version 4o) to generate a hypothesis for 2 entailments, 2 neutrals, and 2 contradictions that use the same premise. Our goal was to expand the space between entailment and contradiction for a given example and by leveraging ChatGPT, we were able to generate and test a lot of examples. The flow-chart below shows how an example is taken through the interface and then generates the 6 outputs for model fine-tuning.

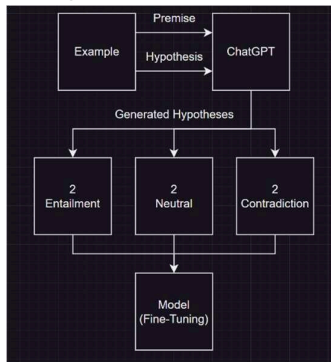


Figure 8: GitHub Workflow Demonstration

The next step is to take the examples generated, for instance for the first example that the model got inaccurate, we now have 6 new training samples that build on top of the same premise. We can now feed them into the model by using the latest checkpoint and to re-evaluate on the new examples.

We repeated this process for the remaining 6 examples (7 in total now) which led to interesting results discussed in the next section.

3 Results

After training the model on the adversarial dataset, the accuracy of the model on the test_r3 split increased from 33% to 41%. This approach led to an increase in accuracy by 8% across the entire test_r3 dataset which means the model can differentiate each class slightly better. The updated accuracy scores below prove this.

Label	Precision	Recall	F1-Score
0	0.41	0.47	0.43
1	0.42	0.51	0.46
2	0.39	0.24	0.30
Accuracy			0.41

Figure 10: Updated Accuracy Scores After Training on ANLI

While the model improved across all classes in precision, recall, and F1-Score, the updated confusion matrix shows that the model is still struggling to learn contradiction samples.

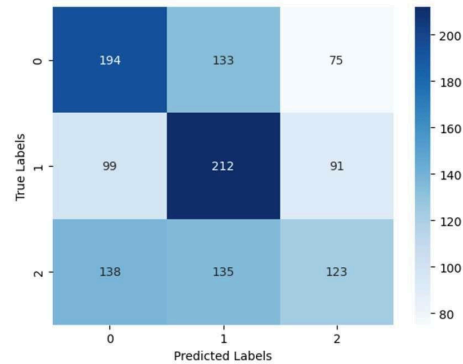


Figure 11: Updated Confusion Matrix After Training on ANLI

After training the model with our newly augmented data that was generated by ChatGPT and upon model analysis against the Contrastive Set, we have found a positive trend for the model in terms of its accuracy.

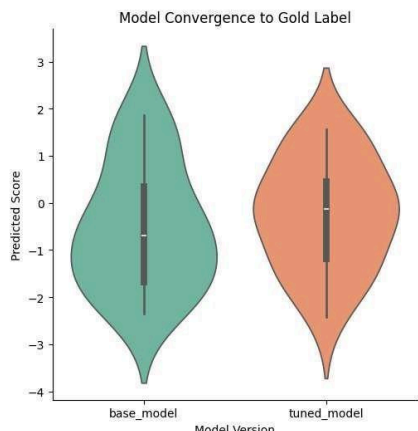


Figure 9: Model Convergence to Gold Label, (Green) old model, (Orange) fine-tuned model with new examples.

The figure above shows the model before training (Green) and the model after we augmented the new data to support the premises (Orange). The important metric here is that the y-axis resembles the label score predicted given the gold label, in our case the higher the mean (White dash in the middle), the higher the score / accuracy for the correct label. With careful examination, we can see that the newly trained model is trending positively in favor of the gold label and thus translating to the model improving for the subset of examples we chose. In numbers, the average score (The white dash in each model in Figure 9 above) for predicting the correct label improved from 0.52 to -0.24 after introducing the generated samples to the model for our examples. The accuracy on the contrastive set before training was 33% and after introducing the newly generated examples it went up to 55% with a positive trend of scores becoming closer to the gold label.

Based on the 7 targeted samples in section 2.2, these data augmentations have proven that the model can and will learn from future samples that specifically target the weaknesses we found. With our current approach, we believe that if we apply this methodology to many more examples, we can

then fine-tune the model and improve its overall accuracy. Sometimes the model may get a lower accuracy than speculated but when we examine specific metrics such as the label score compared to the true label, it is easily visible that the model is learning from the new examples and applying that knowledge to the entire dataset.

4 Conclusion

Models can unintentionally learn dataset artifacts to predict labels, instead of learning the classification task. The specific patterns it learns about the dataset allows it to derive high evaluation performance on the in-distribution dataset.

The issue with this however, is that the model will not generalize well to datasets that don't exhibit the learned patterns. We proved this by training ELECTRA-small on the SNLI dataset and then evaluating on one of the ANLI test splits. The result was poor performance on the ANLI dataset, since the adversarial dataset did not have the same patterns learned from the SNLI dataset. Upon seeing the poor performance, we analyzed the errors and found that the model was doing poor across all label types: entailment, neutral and contradiction but was doing especially bad on contradiction examples. Furthermore, the model did not do well on specific types of examples where it did not understand the context of either the premise or the hypothesis.

The approaches we made to improve the model's performance on the ANLI test dataset did not substantially improve the model's performance. However, the approaches set it on the right track to be improved even further. After the first approach, the model was able to learn better associations and patterns which enabled it to increase its accuracy by 8%. However, the model was still struggling to properly separate the embedding spaces for each label. We then curated contrastive data sets using Chat-GPT to further improve the model's ability to differentiate between each class. The contrastive sets improved accuracy by a small amount, but we speculate that creating a larger contrastive set would help the model even more with separating the embedding spaces of the labels.

It is apparent that the data that gets fed into the model is crucial to determining whether the model

will learn the task. When curating a dataset, we need to ensure that the data is not biased and does not exhibit extreme patterns in one label class but not the others. Furthermore, we need to ensure that the number of samples for each label should be equal to one another, so that the model does not have bias towards one label over another. By completing these requirements, we make it viable for the model to learn the actual task. As a sanity check, it's important to still verify whether the model is learning by testing it using an approach such as evaluating on Adversarial Sets. Furthermore, it may be beneficial to introduce Contrastive Sets during the evaluation phase of the model to identify and improve contextual weaknesses against specific domains.

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